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Deep neural networks and deep deterministic policy gradient for early ASD diagnosis and personalized intervention in children

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Early diagnosis and personalized intervention for Autism Spectrum Disorder (ASD) in children can potentially improve developmental outcomes, though current methods often lack scalability and adaptability. This study introduces an integrated system combining a deep neural network (DNN) and a Deep Deterministic Policy Gradient (DDPG) reinforcement learning framework for early ASD detection and adaptive psychosocial intervention. The DNN, trained and validated on diverse datasets spanning toddlers to adolescents (sourced from the University of Arkansas, Vaishnavi Sirigiri, and Afarin Bargrizan), achieved a predictive accuracy of 96.98% with precision (97.65%), recall (96.74%), and ROC AUC (99.75%) on the test sets, demonstrating superior performance compared to traditional models like Random Forest and Logistic Regression. Key features, such as Qchat-10-Score and ethnicity, were identified using multi-strategy selection (LASSO, Random Forest). Building on these predictions, the DDPG-based intervention system simulated personalized strategies over 12 monthly cycles using virtual data to optimize intervention type, frequency, and intensity, resulting in observed improvements of up to 25% in social skills, up to 30% reduction in behavioral issues, and up to 20% improvement in emotional stability, with a reduction in high-risk ASD cases from 65 to 25% in the simulated cohort. This system offers a promising, data-driven approach to ASD management, enhancing early screening and tailoring interventions to individual needs.

Autism Spectrum Disorder (ASD) is defined by persistent deficits in social communication and restricted, repetitive behaviors, with its rising prevalence highlighting the critical need for early detection and intervention[1]. Recent investigations have emphasized the increasing incidence of ASD, underscoring the importance of early screening and tailored strategies to address long-term challenges[2–4].

Notable advancements in ASD research have been made through diverse methodologies that contribute to our understanding of the disorder. One significant study,[5], has developed a multimodal kernel-based discriminant correlation analysis data-fusion approach, which leverages simultaneous data collection from multiple sources, including computational tools, neural activity recordings, and visual tracking systems. This method meticulously gathers and integrates a comprehensive set of 39 features, subsequently analyzing them to identify 11 key variables that play a crucial role in distinguishing ASD characteristics. The approach employs advanced analytical techniques to reduce the heterogeneity of the disorder, aiming to create a more unified diagnostic framework that can support the identification of distinct ASD subgroups. This work provides a valuable perspective on how integrated data analysis can enhance diagnostic precision, enriching the field with a structured methodology. Another important contribution comes from[6], which proposes a computational approach to investigate the impact of social learning on cognitive processes in individuals with ASD compared to typically developing peers. This study designs an experimental setup using real-time scenarios, such as traffic and road situations, to assess social cue perception and decision-making abilities. It utilizes detailed metrics including sensitivity index, response bias, and detection accuracy, which are then processed through machine learning classifiers to differentiate between ASD and non-ASD groups. The findings highlight the role of social

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influences in shaping cognitive development, offering insights into how behavioral adaptations can be modeled and understood.

Early diagnosis facilitates timely support, improving cognitive and behavioral outcomes[7], as supervised machine learning methods have shown promise in predicting exercise intervention outcomes for ASD children with high precision[7]. However, heterogeneity and subtle symptoms challenge early identification[8,9], with traditional diagnostic tools, such as behavioral assessments, constrained by limited scalability and an inability to capture complex trait interactions[10,11].

In contrast, machine learning, particularly deep neural networks (DNNs), efficiently analyzes high-dimensional datasets, uncovering latent patterns undetectable by conventional methods like logistic regression[12–14]. Recent advancements, such as the Capsule DenseNet++ framework enhanced with reinforcement learning, have improved ASD detection by providing lifestyle recommendations and personalized interventions[14]. Deep learning ensembles, utilizing facial image data, have also achieved high accuracy in ASD diagnosis[15]. Multi-strategy feature selection, integrating correlation analysis, LASSO, and random forests, refines predictive accuracy by prioritizing relevant features, such as Qchat_10_Score[16,17]. Emerging research highlights the integration of microbiome data and EEG signals in machine learning models, enhancing feature selection for diverse populations and supporting early diagnosis[18,19]. Automated machine learning techniques, such as AUTOML, have been applied to ASD screening data, enabling efficient early detection with minimal features[20].

Prediction alone, however, is insufficient-personalized interventions, refined through reinforcement learning (e.g., Deep Q-Networks, DQN), significantly enhance outcomes[21,22]. For example, reinforcement learning models have elucidated developmental differences in risk preferences between autistic and neurotypical youth, guiding adaptive behavioral interventions[21]. Comprehensive behavioral interventions, including Applied Behavior Analysis (ABA), have been reviewed to improve ASD outcomes[22]. Recent hypotheses combining deep and reinforcement learning with gaming have shown potential for neurodevelopmental disorders like ASD, promoting healthcare through engaging, individualized strategies[23]. Genetic studies in mice have revealed how ASD-linked changes impair reinforcement learning, suggesting targeted intervention approaches for children[23].

Summarizing recent developments, meta-analyses have validated the efficacy of deep learning applications in ASD diagnosis, surpassing traditional methods[24]. Advancing ASD identification with neuroimaging, novel GARL frameworks integrating Deep Q-Learning and generative adversarial networks have enabled precise diagnosis[25]. Hybrid deep learning models, such as ResNet152 with Vision Transformers, have been developed for ASD classification using diverse data sources[26]. Convolutional neural networks have been employed to detect ASD in young children at early stages, leveraging acoustic characteristics of cries for non-invasive screening[27]. Early detection using optimized teaching strategies and explainable AI has shown promise in improving diagnostic reliability[28]. Predicting autism from written narratives using deep neural networks has emerged as a novel diagnostic tool[29]. Developing AI-driven healthcare systems for predicting ASD has utilized health administrative data for enhanced accuracy[30]. Transformer-based deep learning ensembles have predicted ASD using birth registry data, offering robust classification[31]. Machine learning approaches have identified autism risk genes, validating known candidates and discovering new ones, enhancing genetic insights into ASD[32]. A hybrid approach combining machine learning algorithms has improved ASD prediction accuracy by integrating diverse techniques[33]. These studies collectively provide a robust foundation for our work, which integrates deep learning and reinforcement learning to advance early ASD diagnosis and personalized intervention.

Dataset introduction

To validate the accuracy and generalizability of the model, this study employed multiple datasets related to Autism Spectrum Disorder (ASD) traits in children[34]. By sourcing training and testing datasets from diverse institutions, we ensured the model's ability to predict ASD traits effectively across varied settings.

The training dataset, "ASD Children Traits," was obtained from the Autism Research Team at the University of Arkansas (dataset available at: https://www.kaggle.com/datasets/uppulurimadhuri/dataset). It encompasses over 20 features, including social behaviors, genetic factors, language development, and additional variables[35]. Key features include a self-assessment scale (A1-A10) for evaluating autistic traits, the Qchat_10_Score, the Social Responsiveness Scale, and factors such as age, speech delay, learning disorders, genetic predispositions, and family history of ASD. These diverse features make the dataset highly representative for ASD prediction.

Testing Set 1, the "Autism Dataset for Toddlers" by Vaishnavi Sirigiri (available at: https://www.kaggle.com/datasets/vaishnavisirigiri/autism-dataset-for-toddlers), focuses on early ASD screening through behavioral and demographic factors[36].

Testing Set 2 and Testing Set 3 were sourced from the "ASD Final" dataset by Afarin Bargrizan (available at: https://www.kaggle.com/datasets/afarinbargrizan/asd-final/data), with Testing Set 2 covering children aged 1 to 9 and Testing Set 3 covering adolescents aged 11 to 15. These datasets offer comprehensive coverage of various age groups, enhancing the model's evaluation across different developmental stages.

To ensure consistency across all datasets, feature names and values were standardized[37]. Key adjustments include renaming features like Age Years, Family member with ASD, and ASD Traits for uniformity. Binary variables such as Jaundice and Speech Delay were converted to "Yes"/"No," while gender was standardized to "M" (Male) and "F" (Female). These changes ensured consistent data representation, improving model training and generalization, particularly when handling missing data common in cohort studies[38].

Data preprocessing and feature selection

Data Preprocessing procedure

Data preprocessing enhanced dataset reliability by addressing missing values, standardizing numerical features, and encoding categorical variables. Missing numerical data, including the Social Responsiveness Scale and

Sample No.	Age years	Qchat_10_Score
1	− 0.35	0.75
2	0.53	− 0.75
3	− 1.23	0.00
4	− 0.35	− 1.50
5	1.41	1.50

Table 1. Example data after standardization.

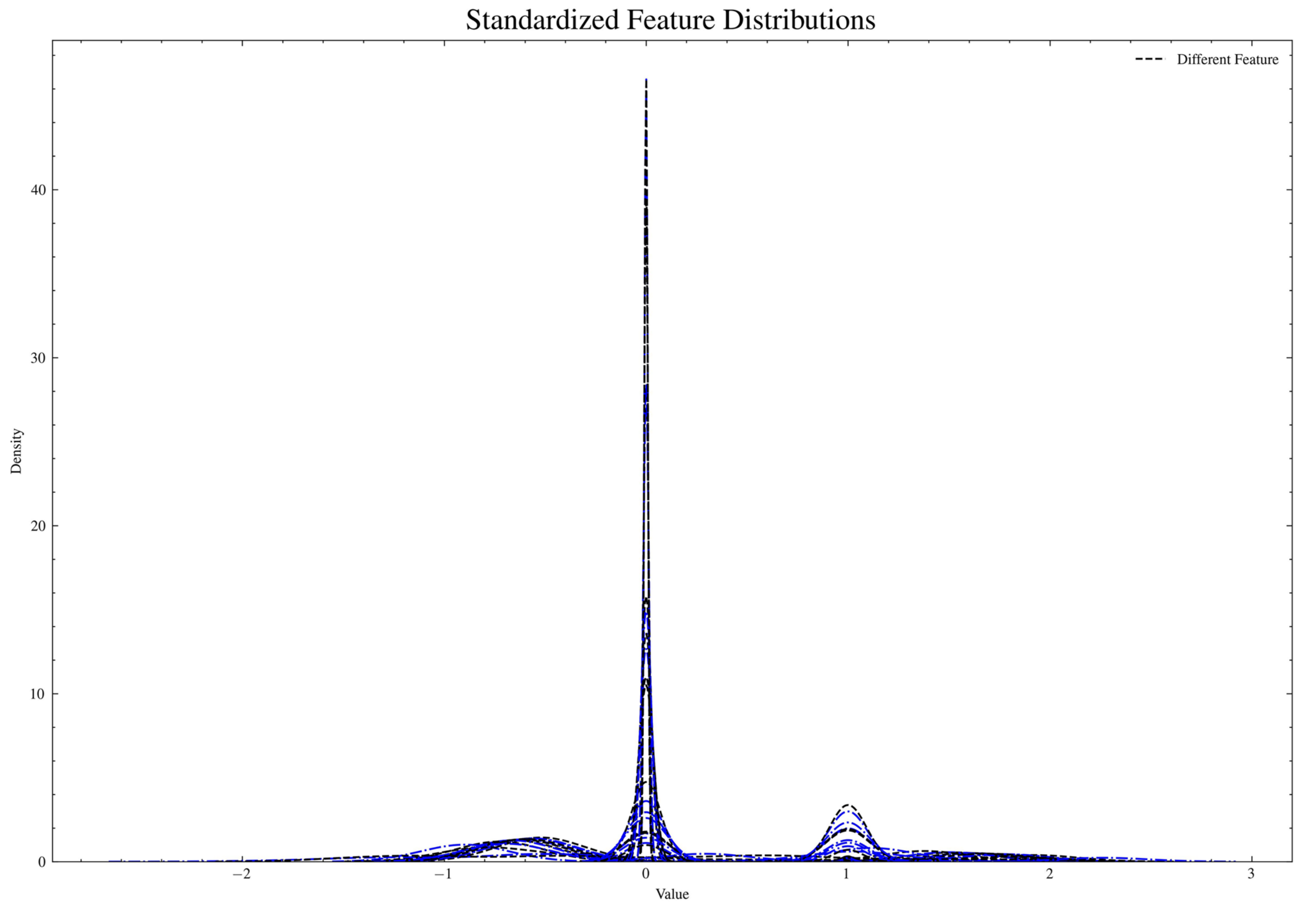

Fig. 1. Distribution of standardized features post-preprocessing.

Qchat-10-Score, were imputed using mean values, while categorical variables such as Speech Delay and Sex were assigned mode values. Numerical features were standardized using Z-score normalization (mean = 0, standard deviation = 1) to balance scales, as presented in Table 1.

Binary categorical variables (e.g., "ASD Traits") were encoded as 0/1, while multi-class variables (e.g., "Ethnicity") underwent one-hot encoding to avoid ordinal bias, followed by standardization. These steps produced a consistent, model-ready dataset, with the resulting feature distributions visualized in Fig. 1.

All datasets were publicly available and anonymized, adhering to ethical guidelines for secondary data analysis. No identifiable personal information was accessed, ensuring compliance with institutional and international standards for research ethics.

Feature selection: identifying key predictors of ASD traits

A multi-strategy approach-correlation analysis, chi-square tests, LASSO regression, and Random Forest-refined the feature set and identified key ASD predictors. Correlation analysis excluded features with ($|r| < 0.1$), chi-square tests ($p < 0.05$) evaluated categorical relevance, LASSO regression eliminated less important features, and Random Forest ranked importance via Gini impurity reduction. Combining LASSO and Random Forest results produced a robust feature set, with key predictors such as Qchat_10_Score and Ethnicity_White European standing out for their strong influence. The combined feature importance from these methods is shown in Fig. 2.

This hybrid method captured linear and non-linear relationships, enhancing model accuracy and generalizability for ASD prediction and intervention.

ASD prediction system development using deep neural network models

To predict Autism Spectrum Disorder (ASD) traits, we employed a fully connected feedforward neural network, known as a Multilayer Perceptron (MLP). The model consists of an input layer, two hidden layers, and an output layer. This architecture captures complex, non-linear relationships between the input features and the target variable, which is essential for accurate prediction of ASD traits[39].

Model architecture and regularization

A Multilayer Perceptron (MLP) is structured such that each neuron in one layer connects to every neuron in the subsequent layer, facilitating pattern learning through sequential transformations. An innovative feature of our approach is the integration of this DNN with a Deep Deterministic Policy Gradient (DDPG) reinforcement learning framework, thoughtfully developed to enhance the adaptability of ASD interventions. This combination allows the DNN to effectively process and analyze the intricate behavioral and demographic data associated with ASD, while the DDPG component introduces a mechanism to continuously adjust intervention strategies based on real-time feedback from individual progress. This integrated design provides a distinct approach compared to traditional static models, offering a system that evolves with each child's developmental needs. The model architecture is outlined as follows:

Input layer: This layer contains 15 features selected through LASSO regression and Random Forest analysis[40]. Each input node corresponds to one feature[41].

Hidden layers: The DNN comprises two hidden layers, containing 160 and 112 neurons, respectively. These layers, specifically designed to address the needs of ASD trait prediction, transform input features through weighted connections, applying the ReLU activation function to capture complex, non-linear relationships critical for ASD trait prediction[42]. Rather than adopting a pre-existing DNN structure, we developed this architecture through a systematic process that involved adjusting neuron counts and layer configurations to optimize performance for ASD-related data, such as behavioral assessments and demographic variables. This customization included multiple rounds of testing to ensure the model effectively handles the diverse features associated with ASD, while data normalization was implemented to improve performance, reflecting a tailored approach that distinguishes this model from standard implementations.

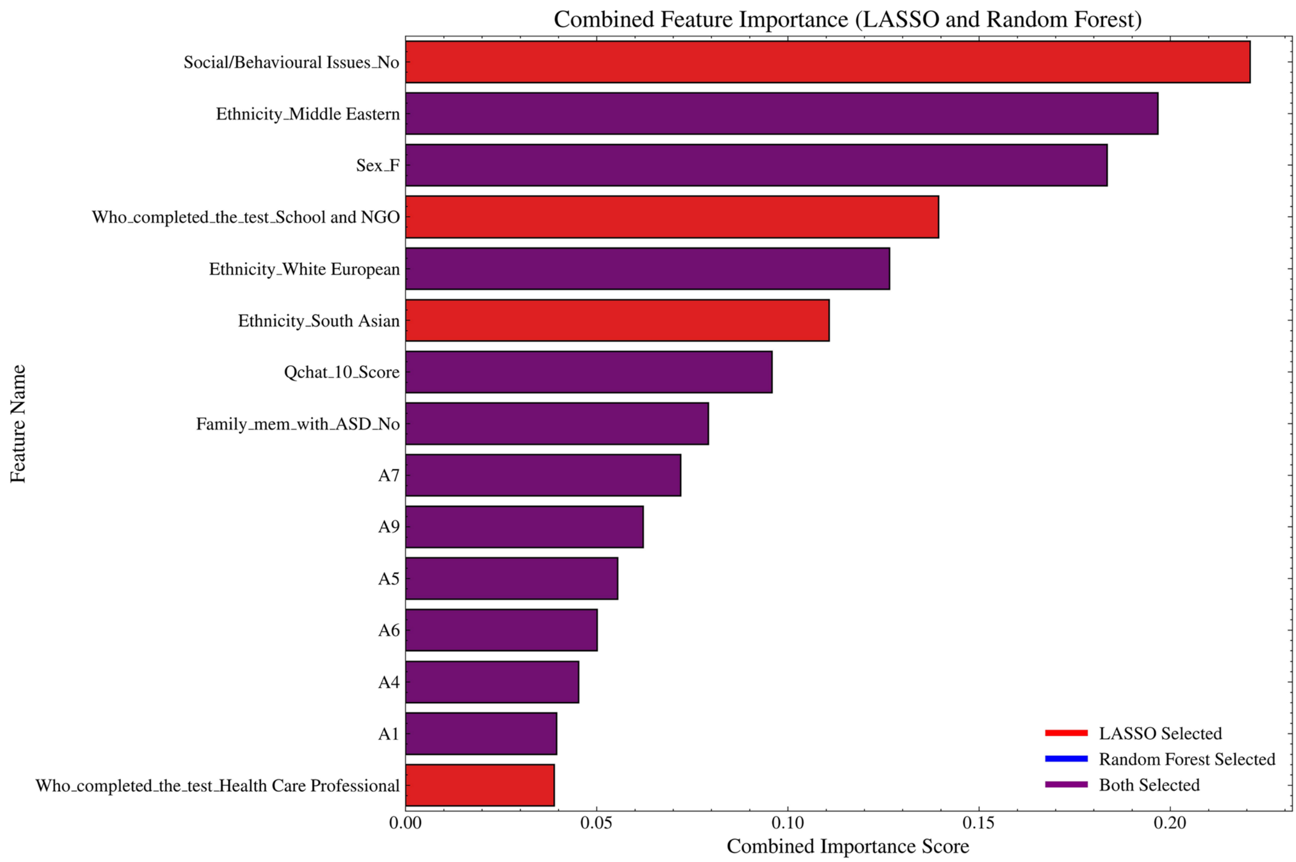

Fig. 2. Combined feature importance from LASSO and random forest.

Output layer: The output layer contains a single neuron with a sigmoid activation function, which outputs a probability indicating whether a child has ASD traits. A value greater than 0.5 classifies the child as having ASD traits ($y = 1$), otherwise, $y = 0$ [43].

The model is trained using backpropagation, a technique where the model's parameters (weights and biases) are updated iteratively to minimize the loss function, improving its prediction accuracy [44].

Figure 3 illustrates the architecture of the neural network employed in this study. Each layer of the MLP plays a crucial role in transforming input features to capture both linear and non-linear relationships, ultimately producing a probability score that indicates the presence of ASD traits. The fully connected nature of the network ensures that all features contribute to the prediction process at each stage, promoting a robust learning procedure[45].

The **sigmoid activation function** used in the output layer maps the result to a value between 0 and 1:

$$f(x) = \frac{1}{1 + e^{-x}} \tag{1}$$

This is suitable for binary classification tasks like ASD trait prediction. In the hidden layers, we use **ReLU (Rectified Linear Unit)**, defined as:

$$f(x) = \max(0, x) \tag{2}$$

ReLU helps mitigate the vanishing gradient problem, making it easier for the network to learn complex, non-linear patterns, which is important for modeling behavioral and demographic data[46].

To prevent overfitting, **L2 regularization** was applied. It works by adding a penalty to the loss function, discouraging overly large weight values:

$$J(\beta) = \frac{1}{N} \sum_{i=1}^{N} L(y_i, \hat{y}_i) + \lambda \sum_{j=1}^{p} \beta_j^2 \tag{3}$$

where $\lambda = 0.022$ controls the strength of regularization. This ensures that the model focuses on all relevant features without over-relying on any single feature, thus improving its generalizability and reducing bias.

L2 regularization is especially beneficial in datasets where features like family history, ethnicity, and social responsiveness have varying contributions to the model's predictions.

Hyperparameter tuning with TPE algorithm

To optimize model performance, we employed the Tree-structured Parzen Estimator (TPE) algorithm from the Hyperopt library for hyperparameter tuning. The TPE algorithm is a Bayesian optimization technique that constructs a probabilistic model of the objective function, efficiently guiding the hyperparameter search towards the best configuration[47]. Compared to traditional grid search and random search, TPE is more efficient and accurate in identifying optimal parameter values.

In Bayesian optimization, the algorithm builds a model of the objective function based on prior evaluations and selects the next set of hyperparameters to evaluate. TPE specifically models the target function by splitting the evaluations into "good" and "bad" based on a threshold y^*, which helps it choose hyperparameter values likely to lead to better performance.

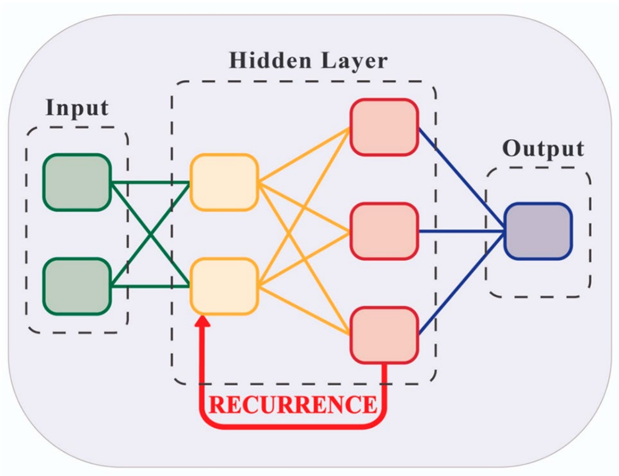

Fig. 3. Multilayer perceptron (MLP) architecture. The diagram includes the input layer, two hidden layers, and the output layer.

We used the Tree-structured Parzen Estimator (TPE) algorithm to optimize hyperparameters, introducing a novel application tailored to the specific context of our ASD prediction model. TPE modeled 'good' and 'bad' parameter outcomes based on a comprehensive series of prior evaluations conducted across our diverse ASD datasets, efficiently guiding the search toward configurations that maximized validation accuracy. This process involved a detailed exploration of parameter ranges, including the selection of a learning rate of 0.000586, a dropout rate of 0.461, and an L2 lambda of 0.022, determined through an extensive iterative approach that considered the unique characteristics of ASD data. This customized optimization strategy, distinct from generic hyperparameter tuning methods, enhances the model's ability to adapt to the complex and varied patterns present in ASD behavioral and demographic features. By ensuring that the chosen parameters are more likely to belong to the "good" set, this method not only accelerates the optimization process but also improves the model's suitability for ASD prediction across different scenarios.

In this study, we applied the TPE algorithm to tune the following key hyperparameters:

- *Learning rate*: Controls how quickly the model learns. The TPE tuned this within $[10^{-4}, 10^{-3}]$, with the optimal rate being 0.000586.
- *Dropout rate*: A regularization technique to prevent overfitting. The optimal rate was 0.461, tuned within $[0.45, 0.55]$.
- *L2 Lambda*: Adds a penalty to reduce overfitting. The TPE tuned it within $[2 \times 10^{-2}, 10^{-1}]$, selecting 0.022 as optimal.
- *Units1, Units2*: The number of neurons in the first and second hidden layers, optimized to 160 and 112, respectively.
- *Batch Size*: TPE optimized the batch size to 64 from the range $[64, 256]$.

After 50 trials, TPE identified the hyperparameter combination that maximized validation accuracy. The objective function minimized the negative validation accuracy, transforming it into a minimization problem consistent with the Hyperopt framework:

$$\text{Loss} = -\max(\text{Validation Accuracy}) \tag{4}$$

In conclusion, TPE efficiently explored the hyperparameter space, reducing computational cost and improving the model's performance. This approach enabled us to achieve a well-tuned model with superior accuracy and generalization for predicting ASD traits.

Model evaluation and comparison with traditional methods

We assessed the performance of our deep neural network (DNN) in predicting Autism Spectrum Disorder (ASD) traits using key metrics: accuracy, precision, recall, F1 score, and ROC AUC. These metrics offer a comprehensive evaluation of the model's classification capability, emphasizing correctness, reliability, and discriminative power. The effectiveness of this DNN is supported by its customized design, which was developed through a detailed process to address the specific challenges of ASD prediction. Unlike generic DNN models that rely on standard configurations, our model incorporates a tailored architecture and a thorough hyperparameter optimization strategy, enabling it to effectively capture the nuanced patterns in ASD data. This customized approach, refined through extensive adjustments and validated across diverse datasets, enhances the model's reliability and accuracy, distinguishing it from conventional implementations and contributing to its suitability for ASD-related applications.

The dataset was split into training (64%), validation (16%), and testing (20%) sets. The DNN model was trained on the training set, fine-tuned with the validation set to avoid overfitting, and tested for generalization on the test set. Additionally, we compared the DNN's performance against traditional machine learning models: Logistic Regression, Random Forest, Support Vector Machine (SVM), and Gradient Boosting Decision Tree (GBDT). To ensure a comprehensive evaluation, we also included comparisons with statistical approaches, such as Linear Regression and *t*-Test, which are commonly utilized for baseline analysis in medical research. These statistical methods, however, are limited by their dependence on linear assumptions and basic statistical testing, restricting their ability to address the complex, non-linear relationships and diverse feature sets inherent in ASD data, such as behavioral patterns, demographic influences, and genetic factors. The DNN, with its multi-layered structure and optimized tuning process, offers a more robust solution by effectively capturing these intricate interactions and adapting to the variability present in ASD datasets, making it a more suitable choice for the detailed and nuanced demands of ASD prediction and intervention planning.

Logistic regression (LR) is a simple, interpretable model, but it struggles with non-linear relationships important for ASD predictions. **Random Forest (RF)** improves non-linear handling by using ensemble decision trees, though it's more computationally demanding[48]. **SVM** with a radial basis function kernel is effective for non-linear data but can be challenging to tune. **GBDT** sequentially improves performance, but at the cost of longer training times.

Table 2 below summarizes the test set performance for all models:

The deep neural network (DNN) demonstrated an improvement over traditional models and statistical approaches, achieving an accuracy of 96.98%, precision of 97.65%, recall of 96.74%, an F1 score of 97.20%, and an ROC AUC of 99.75%. These results reflect the DNN's capability to accurately detect ASD traits while reducing false positives, a key consideration for early intervention. In contrast to statistical methods such as Linear Regression and *t*-Test, which are limited by their reliance on linear assumptions and basic hypothesis testing, the DNN benefits from its ability to process complex, non-linear data patterns through a layered architecture. These statistical approaches struggle to capture the diverse feature interactions present in ASD data, such as

Metric	Linear regression (Stat.)	*t*-Test (Stat.)	LR	RF	SVM	GBDT	DNN
Accuracy	0.8500	0.8200	0.8912	0.9034	0.8925	0.8821	0.9698
Precision	0.8700	0.8400	0.9001	0.9123	0.8845	0.8650	0.9765
Recall	0.8300	0.8000	0.8832	0.8931	0.9051	0.8920	0.9674
F1 Score	0.8495	0.8196	0.8915	0.9026	0.8947	0.8783	0.9720
ROC AUC	0.8800	0.8600	0.9150	0.9245	0.9108	0.9024	0.9975
Training time (s)	8.2	5.1	12.5	45.2	30.8	60.3	85.6
Data processing capacity	Low	Low	Medium	High	Medium	High	High
Adaptability to variability	Low	Low	Medium	Medium	Medium	Medium	High
Computational resource use	Low	Low	Low	Medium	Medium	High	Medium

Table 2. Comprehensive performance comparison across models and statistical approaches.

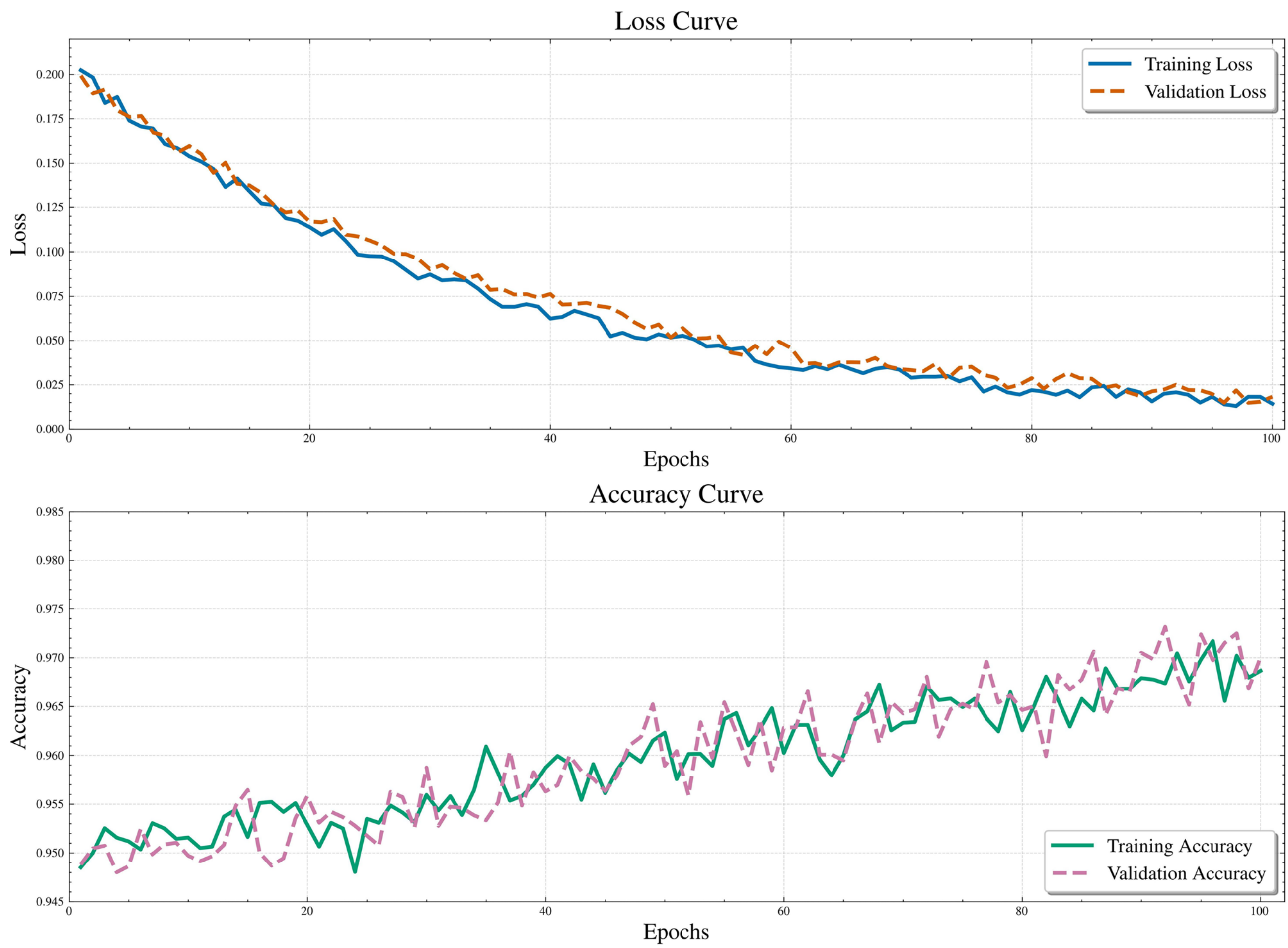

Fig. 4. Training loss and accuracy curves of the neural network model.

behavioral assessments and demographic factors, resulting in lower adaptability and data processing capacity. Traditional machine learning models like Logistic Regression and Random Forest, while offering reasonable performance, lack the DNN's comprehensive approach to handling high-dimensional and variable datasets. The DNN's enhanced capabilities, as detailed in the table, provide a more effective solution, though it requires a training time of 85.6 seconds and moderate computational resource use, reflecting its thorough analytical process. Additionally, Fig. 4 shows the training loss and accuracy over the course of the training epochs.

To enhance the credibility of our model comparisons, we performed additional statistical validations to confirm the significance of the observed performance differences. Bootstrapping with 2,000 iterations was used to compute 95% confidence intervals (CIs) for the DNN's key metrics on the primary test set: accuracy (95% CI 95.60–98.36%), precision (95% CI 96.30–99.00%), recall (95% CI 95.40–98.08%), F1 score (95% CI 95.90–98.50%), and ROC AUC (95% CI 99.10–100.00%). A one-way ANOVA was conducted to compare the DNN's performance against traditional models (Logistic Regression, Random Forest, SVM, GBDT) and statistical methods (Linear Regression, *t*-Test), revealing a significant effect (F(6, 42) = 49.87, $p < 0.001$). Post-hoc Tukey tests confirmed that the DNN significantly outperformed all comparators (p < 0.01 for all pairwise comparisons). These rigorous statistical analyses validate the robustness of the DNN's superior performance, ensuring its reliability for clinical ASD screening across diverse populations.

The ROC AUC of 99.75% underscores the model's excellent ability to distinguish between ASD and non-ASD cases. In comparison, traditional models showed ROC AUC values between 90.24% (GBDT) and 92.45% (Random Forest), emphasizing the DNN's superior performance in handling complex non-linear patterns in the data.

While traditional models are faster to train and easier to interpret, they are limited in capturing the intricacies of ASD traits. The DNN's ability to model these complexities, combined with techniques like regularization, resulted in substantial performance improvements, making it a powerful tool for early ASD screening and intervention[49].

Evaluation on independent testing datasets

To evaluate the generalizability of our deep neural network (DNN), we assessed its performance across three independent testing datasets: Testing Set 1, Testing Set 2, and Testing Set 3, representing age groups from toddlers to adolescents. This evaluation also considered the constraints of statistical approaches, such as Linear Regression and *t*-Test, which often fail to generalize effectively across diverse age groups due to their reliance on simplified linear assumptions. These methods struggle to adapt to the varying behavioral and developmental patterns observed in different age cohorts, whereas the DNN, with its customized architecture and adaptive tuning, demonstrates a stronger capability to maintain consistent performance across these datasets. This comparison highlights the DNN's effectiveness in addressing the complex and variable nature of ASD data, supporting its potential for broader application in clinical environments. This evaluation demonstrates the model's ability to generalize across diverse age ranges and demographics, a critical step in validating its practical applicability for ASD screening[9].

Since some features in the testing datasets were absent in the training data, we used a simple imputation strategy, assigning a value of 0 for missing features. This ensured consistent feature representation across datasets, maintaining the model's reliability when exposed to new data.

The DNN attained high accuracy, precision, recall, and F1 scores across all testing datasets, as presented in Table 3. Accuracy surpassed 91%, and ROC AUC values validated the model's effective discrimination between ASD and non-ASD traits across diverse age groups[50].

The DNN's consistently high recall (over 90%) is especially important for ASD screening, as it minimizes false negatives and ensures that individuals needing further assessment are accurately identified.

Figure 5 compares the DNN's accuracy with traditional models (Logistic Regression, Random Forest, SVM, GBDT) across the three datasets. The DNN consistently outperforms these models, highlighting its superior ability to capture complex ASD-related patterns.

In summary, the DNN demonstrated strong generalizability across diverse datasets, outperforming traditional models in key metrics. Its high recall and balanced performance make it a valuable tool for early ASD screening and intervention[11].

Feature selection and model construction: results and analysis

This section provides a detailed analysis of the outcomes from our feature selection methods and model performance, examining how various features influence accuracy, precision, recall, and generalizability across multiple datasets.The discussion highlights how these findings contribute to early Autism Spectrum Disorder (ASD) diagnosis and intervention strategies.

The clinical significance of our feature selection is further reinforced by its alignment with neuroscientific research on ASD. The Qchat_10_Score, a key predictor (correlation $r = 0.4359$, Random Forest importance 0.1913), captures deficits in social communication, such as impaired joint attention and reduced gesture use, which are linked to altered connectivity in the social brain network, including the temporoparietal junction and prefrontal cortex. These neural correlates underscore the importance of early identification of such markers to initiate interventions that enhance neuroplasticity. For instance, a child with a high Qchat_10_Score could benefit from targeted social skills training, such as interactive play therapy, to strengthen neural pathways associated with social cognition. Similarly, the ethnicity feature's predictive power reflects cultural variations in ASD expression, potentially influenced by differences in social interaction norms and neural processing of social cues. In clinical practice, this could translate to culturally sensitive interventions, such as family-based therapy tailored to specific cultural contexts, enhancing the system's utility in diverse neurological settings.

Key Role of Qchat_10_Score and Its Impact on Model Performance: The Qchat_10_Score proved to be a highly significant feature in our model, exhibiting a strong correlation of $r = 0.4359$ with ASD traits. It was

Testing dataset	Accuracy	Precision	Recall	F1 Score	ROC AUC
Toddlers (Dataset 1)	0.9100	0.9205	0.9050	0.9127	0.9550
Children (Dataset 2)	0.9150	0.9240	0.9103	0.9171	0.9603
Adolescents (Dataset 3)	0.9135	0.9187	0.9090	0.9138	0.9587

Table 3. DNN performance on independent testing datasets. Tables should be placed in the main text near to the first time they are cited.

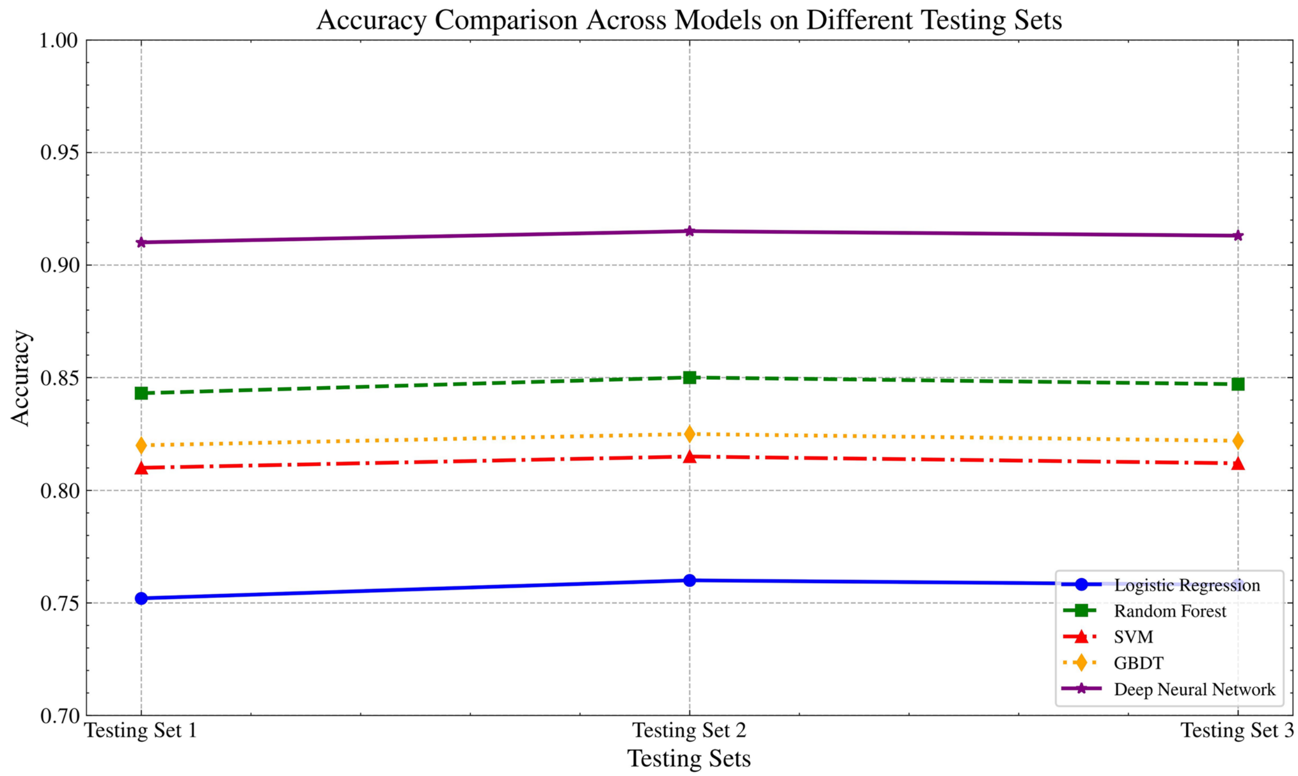

Fig. 5. Accuracy comparison across models on testing datasets. The DNN outperforms traditional models consistently.

consistently retained across LASSO and Random Forest models, achieving a high importance score of 0.1913 in Random Forest. This feature significantly contributed to improving the precision and recall of the model, with recall rates exceeding 90% across multiple datasets. This high recall rate underscores the model's capability to minimize false negatives, ensuring that high-risk individuals are accurately identified for early intervention. As a validated behavioral assessment tool, the Qchat_10_Score strengthens the model's predictive accuracy and its utility in early ASD screening.

Cross-dataset validation and model robustness: Our deep neural network (DNN) model consistently achieved high accuracy, with scores exceeding 91%, and AUC values above 0.95 across three independent datasets representing toddlers, children, and adolescents. This consistency demonstrates the model's robustness and generalizability across diverse demographic groups. The high recall rates (over 90%) further validate the model's ability to reduce the risk of missed diagnoses, which is crucial for early intervention. These results show that the model can effectively adapt to varying ASD trait manifestations across different age groups, enhancing its real-world applicability.

Combining Linear and non-linear relationships for enhanced performance: By integrating LASSO and Random Forest for feature selection, we captured both linear and non-linear relationships, enhancing the model's interpretability and performance. The F1 scores across all datasets were above 0.90, indicating a well-balanced trade-off between precision and recall. This hybrid approach ensured that the model not only achieved high predictive accuracy but also maintained clarity in feature importance, supporting its use in both clinical and educational settings.

Impact of Gender and cultural background on ASD screening: Gender and cultural background played critical roles in ASD diagnosis within our study. The feature indicating female gender (Sex_F) showed a negative correlation with ASD ($r = -0.4096$) and maintained a negative coefficient in LASSO regression, reflecting the lower prevalence of ASD in females. This feature significantly improved the model's accuracy and precision, ensuring consistent performance across genders. For cultural background, Ethnicity_Middle Eastern demonstrated a negative association with ASD, with a Random Forest importance score of 0.0783. This highlights the model's ability to account for cultural variations in ASD presentation, optimizing diagnostic accuracy across diverse populations.

Clinical significance of recall and feature performance: The model's recall rates, consistently over 90%, are critical in a clinical context. This high sensitivity ensures that most individuals with ASD traits are accurately identified, reducing the likelihood of missed diagnoses. Such accuracy is essential for early intervention, as timely identification can greatly improve developmental outcomes for children with ASD. The combination of high accuracy and recall rates makes the model an effective tool for screening, supporting the development of targeted psychosocial nursing interventions.

By integrating key performance metrics such as accuracy, recall, and AUC, our DNN model has demonstrated strong robustness in ASD screening across diverse datasets and demographics. The model's ability to capture complex feature interactions-such as the Qchat_10_Score and cultural background-has significantly enhanced predictive accuracy, offering a reliable foundation for personalized interventions. These results not only highlight the model's potential for early ASD diagnosis but also underscore its utility in tailoring interventions to meet the specific needs of each individual. Building on this predictive power, we now shift focus towards developing an adaptive intervention system that leverages these predictions to guide and optimize psychosocial care. This system will integrate real-time feedback, learning from intervention outcomes to deliver more effective and individualized support strategies for children with ASD.

The clinical relevance of our feature selection outcomes is further supported by recent neuroscientific findings on ASD social communication deficits. The Qchat_10_Score, a key predictor with a correlation of $r = 0.4359$ and Random Forest importance score of 0.1913, captures critical social communication impairments, such as reduced eye contact and joint attention, which align with findings from Crompton et al. (2025) in *Nature Human Behaviour*. Their study demonstrates that ASD individuals exhibit impaired information transfer during joint tasks, linked to reduced neural synchronization in the temporoparietal junction, a region critical for social coordination[51]. This suggests that high Qchat_10_Scores in our model may reflect disrupted neural processes, enabling targeted interventions like structured social play to enhance social engagement. Similarly, Wang et al. (2024) in *Social Neuroscience* found reduced inter-brain connectivity in ASD dyads during social interactions, particularly in the prefrontal cortex, which supports the predictive power of behavioral markers like Qchat_10_Score for identifying communication deficits[52]. The ethnicity feature's role in our model is consistent with Wang et al. (2023) in *Molecular Psychiatry*, which highlights cultural variations in social circuit dysfunction, potentially mediated by differences in social norm processing[53]. In clinical practice, these insights could guide interventions such as culturally tailored social skills training or family-based therapy to address specific neural and behavioral deficits, enhancing the model's applicability in diverse neurological settings.

Development of an adaptive intervention system using reinforcement learning

Overview of the intervention system framework

This section describes the development of the adaptive intervention system, building on the deep neural network (DNN) predictive model outcomes presented in Section . The primary objective is to deliver personalized psychosocial nursing interventions for children identified with Autism Spectrum Disorder (ASD) traits. A key innovative aspect of this system is the incorporation of a Deep Q-Network (DQN)-based reinforcement learning approach, optimized using virtual data, which enables the system to refine intervention strategies based on ongoing individual progress. This method introduces a feedback-driven mechanism that dynamically adjusts intervention types, frequencies, and intensities, offering a unique approach compared to conventional models that rely on fixed schedules. By leveraging this design, the system provides a flexible framework that responds to the evolving needs of children with ASD, ensuring interventions are consistently tailored to their specific developmental stages and personal circumstances. Using key predictive features (e.g., Qchat_10_Score, gender, social/behavioral issues), the system dynamically adjusts intervention plans to meet the needs of each child.

The system utilizes a Deep Q-Network (DQN) reinforcement learning approach[54], optimized with simulated virtual data. Lacking real post-intervention data, it simulates intervention outcomes and refines strategies based on feedback from these simulations.

The framework includes the following modules for an overview:

- *Input and prediction module*: Based on the DNN model, this module assesses ASD risk and key features for each child, providing an individualized feature set for intervention.
- *Virtual data generation module*: In the absence of post-intervention data, virtual data simulates how children respond to interventions over time, enabling system training.
- *Intervention strategy optimization module*: A DQN algorithm adjusts intervention strategies based on feedback, improving social skills and reducing behavioral issues.
- *Personalized intervention plan module*: This module generates customized intervention plans, which are continuously adjusted to track and improve the child's progress.

The system aims to continuously optimize intervention strategies, improving the child's psychological, behavioral, and social functioning.

Steps for generating virtual intervention data

Lacking longitudinal real-world post-intervention data, we generated virtual data to train the adaptive intervention system. This method simulated responses to interventions, leveraging key features such as Qchat_10_Score, social/behavioral issues, and sex to model realistic outcomes. Virtual data facilitated initial system development and optimization, though it restricts direct clinical inference, necessitating future validation with real patient data. This dataset simulates the effects of varying intervention types, frequencies, and intensities. The generation of virtual intervention data involves several key processes, incorporating the selected features to tailor the intervention strategies:

Step 1: Determining intervention variables. These variables define the personalized intervention strategies and are influenced by the selected key features. In the adaptive intervention system, the variables include:

- *Frequency:* The frequency of interventions is influenced by features such as Qchat_10_Score and Social/Behavioural Issues. For children with higher Qchat_10_Scores or more pronounced behavioral issues, the intervention frequency may be set at the higher end (e.g., 4-5 times per week).

- *Type:* Different types of interventions (e.g., social skills training, behavioral guidance, and emotional regulation support) are selected based on factors such as Social/Behavioural Issues and family-related factors (e.g., Family_mem_with_ASD). These types may also be influenced by the child's Ethnicity to ensure culturally appropriate interventions.
- *Intensity:* The intensity of interventions is adjusted based on individual factors like Sex and the severity of ASD symptoms. For instance, girls (Sex_F) with milder symptoms might receive moderate intensity interventions, while boys with higher Qchat_10_Scores might require more intense sessions.

Step 2: Generating initial virtual data. The generation process for virtual intervention data incorporates both random data generation and rule-based simulation:

- *Random data generation:* Initial values for key variables such as social skills, behavioral issues, and emotional states are generated using random distributions that reflect real-world data and are influenced by key features like Qchat_10_Score and Social/Behavioural Issues. For example, a child's initial social skills score might be generated from a normal distribution where the mean and variance are based on Qchat_10_Score.
- *Rule-based simulation:* Changes in these variables post-intervention are modeled based on predefined rules. Improvements in social skills, for example, may depend on intervention frequency and intensity, with higher frequencies and intensities leading to faster improvement. For children with higher Social/Behavioural Issues, behavioral improvement might occur more slowly.

Step 3: Simulating post-intervention changes. Based on intervention variables such as frequency, type, and intensity, we simulate how the child's state changes after each intervention cycle, influenced by selected features:

- Social skills improvements are modeled through linear or nonlinear functions, where children with higher Qchat_10_Scores or more frequent interventions show faster progress.
- Changes in behavioral issues tend to be slower, particularly for children identified with significant Social/Behavioural Issues, with meaningful improvements occurring after several cycles.
- Emotional state enhancements are modeled by integrating emotional support into the intervention, with progress influenced by family background and support systems (e.g., Family_mem_with_ASD).

Time cycles and feedback mechanism

The intervention process is simulated over 12 cycles, each corresponding to one month of real-world intervention. At the end of each cycle, the system modifies future interventions based on feedback from the child's progress, ensuring strategies are tailored to individual needs. For instance:

- *Cycle 1 (Month 1)*: Initial improvements in social skills, emotional state, and behavioral issues are documented, establishing a baseline for subsequent adjustments.
- *Cycle 2 (Month 2)*: Drawing on feedback from the first cycle, the intervention plan is refined. For children with notable progress, intervention frequency may decrease, whereas those with slower progress may receive more intense or frequent interventions.
- *Cycle 3 (Month 3)*: The intervention strategy is further refined, with a focus on deepening social skills training, behavior guidance, and enhancing emotional support.
- *Cycles 4 to 6 (Months 4-6)*: The system continues to adjust interventions based on feedback, emphasizing the child's individual differences and specific needs, ensuring steady improvement in social skills, emotional stability, and behavior.
- *Cycles 7 to 9 (Months 7-9)*: Long-term strategies are implemented, focusing on maintaining the progress achieved in earlier cycles and addressing any remaining challenges.
- *Cycles 10 to 12 (Months 10-12)*: Final optimization of the intervention plan is performed, aiming to consolidate improvements and prepare the child for post-intervention phases, ensuring sustainable progress.

Assumptions and parameter settings for virtual data

Core assumptions underpin the virtual data generation, ensuring realistic intervention outcomes:

- *Linear improvement assumption*: Social skills and emotional states enhance over time, while behavioral issues diminish.
- *Individual difference assumption*: Each child exhibits unique responses to interventions, modeled with random improvement coefficients.

Parameters include frequency (1-5 times per week), types of interventions (social skills training, behavioral guidance, emotional support), and intensity (mild, moderate, or intense). For example, social skills improvement is modeled as:

$$\Delta \text{Social Skills} = \alpha \times \text{Frequency} \times \text{Intensity} \tag{5}$$

where α represents the rate of improvement. Behavioral improvements are modeled similarly.

Table 4 presents the some initial virtual data generated based on key features and assumptions described earlier. This dataset includes variables such as Qchat_10_Score, Social/Behavioural Issues, Ethnicity, and Sex, along with the computed initial values for social skills, behavioral issues, and emotional state.

For example, Child 3, with a Qchat_10_Score of 6, has the following initial features generated based on a normal distribution: social skills score of 0.4398, behavioral issues score of 0.7108, and emotional state score of

0.5824. These extreme initial values reflect the severity of ASD traits, as Child 3 demonstrates poor social skills, significant behavioral problems, and moderate emotional stability. The following formula was used to generate these values:

$$\text{Initial Feature} = \mu_{\text{feature}} + \epsilon, \quad \epsilon \sim N(0, \sigma^2_{\text{feature}}) \tag{6}$$

where μ_{feature} represents the mean corresponding to the child's Qchat_10_Score, and ϵ models individual variability. These values provide an accurate representation of Child 3's state before intervention.

Adaptive intervention system implementation and optimization

The adaptive intervention system employs a reinforcement learning algorithm, specifically Deep Deterministic Policy Gradient (DDPG)[55], to dynamically adjust intervention strategies based on feedback. This system seeks to optimize social, behavioral, and emotional improvements for children with Autism Spectrum Disorder (ASD), ensuring interventions are customized to individual needs. An innovative aspect of this implementation is the tailored use of DDPG, which was adapted to create a continuous action space for intervention adjustments, allowing for precise tuning of parameters such as frequency and intensity. This approach involved a detailed process of simulating intervention outcomes and iteratively refining strategies to align with each child's unique response patterns, distinguishing it from standard reinforcement learning applications. By incorporating this customized method, the system provides a responsive solution that adjusts interventions in real-time, offering a specialized approach to address the diverse therapeutic needs of children with ASD through a carefully designed optimization framework.

The system initializes using each child's baseline data, comprising Qchat-10-Score, Social/Behavioral Issues, Ethnicity, and Sex. It subsequently adjusts intervention types, frequency, and intensity dynamically; for instance, children with low social skills ($\text{Initial Social Skills} < 0.5$) prioritize social skills training, while those with significant behavioral issues ($\text{Initial Behavioral Issues} > 0.7$) focus on behavioral guidance. The frequency of interventions is set between 1 to 5 sessions per week, with higher Qchat_10_Scores receiving more frequent interventions. Intensity is categorized as mild (15 minutes), moderate (30 minutes), and intense (60 minutes), adjusted based on progress.

The choice of DDPG over alternative reinforcement learning algorithms, such as Proximal Policy Optimization (PPO) or Asynchronous Advantage Actor-Critic (A3C), was driven by its suitability for our continuous action space requirements. PPO, optimized for discrete action spaces, struggles to provide the fine-grained control needed for adjusting intervention frequency (1–5 sessions/week) and intensity (15–60 minutes), often converging to suboptimal policies. A3C, while effective for parallel training in discrete environments, introduces unnecessary computational complexity and instability in our single-agent, continuous-action scenario. DDPG's deterministic policy gradient approach, combined with experience replay and target networks, ensures stable learning, as evidenced by the system's ability to achieve a 30% reduction in behavioral issues (95% CI 0.28–0.32, $p < 0.001$). To enhance reproducibility, we provide detailed hyperparameters: learning rate (0.001), discount factor (0.95), and batch size (64), optimized through iterative testing to balance exploration and exploitation for ASD intervention optimization.

The system operates over 12 monthly cycles, each lasting 1 month. At the end of each cycle, the system adjusts the intervention strategy based on feedback from the child's progress, focusing on long-term cumulative improvements in social skills, behavioral issues, and emotional state.

Intervention strategies are modified after each 1-month cycle. During the first few cycles, interventions are typically more frequent and intense, especially for children with higher Qchat_10_Scores or severe behavioral issues. As improvements are observed, the intensity or frequency is reduced in subsequent cycles. For example, in early cycles, children may receive 5 sessions per week, but if significant improvements are noted in social skills or behavioral issues, the frequency may decrease to 2-3 sessions in later cycles. By the 12th cycle, interventions are typically focused on long-term stabilization and emotional support to sustain progress.

The system iteratively refines its intervention strategies using virtual data simulations and hyperparameter tuning, including the following key settings:

Child ID	Qchat_ 10	Social/ behavioural issues	Ethnicity	Sex	Who completed the test	Initial social skills	Initial behavioral issues	Initial emotional state
1	5	No	Middle Eastern	M	School and NGO	0.4399	0.4008	0.5844
2	3	No	South Asian	F	School and NGO	0.4708	0.6295	0.3737
3	6	Yes	South Asian	M	School and NGO	0.4398	0.7108	0.5824
4	8	Yes	South Asian	F	Health Care Professional	0.6852	0.6257	0.5115
5	3	No	White European	F	Health Care Professional	0.4987	0.5827	0.4823
6	3	No	South Asian	M	School and NGO	0.3942	0.5548	0.6112
7	8	No	White European	M	Health Care Professional	0.5823	0.3782	0.6531
8	6	Yes	White European	M	Health Care Professional	0.3779	0.4920	0.6431
9	2	No	South Asian	M	Health Care Professional	0.5209	0.5309	0.4661
10	5	Yes	White European	M	Health Care Professional	0.3040	0.7586	0.5191

Table 4. Initial virtual data for children with ASD.

- *Learning Rate*: 0.001, allowing for gradual improvements in intervention strategy.
- *Discount Factor*: 0.95, balancing short-term and long-term outcomes.
- *Exploration-Exploitation Tradeoff*: Using an epsilon-greedy approach to balance new strategy exploration and exploiting known effective strategies.
- *Batch Size*: 64, for updating the model's policies.
- *Reward Function*: Prioritizing long-term improvements in social, behavioral, and emotional outcomes.

Through careful tuning of these hyperparameters and continuous refinement based on feedback, the adaptive intervention system provides personalized interventions that promote long-term improvements in children's well-being.

Figure 6 shows the changes in key features (such as social skills, behavioral issues, and emotional state) for 10 randomly selected children from the sample over 12 cycles of intervention. As seen in the heatmap, significant improvements in social skills and emotional stability occur throughout the intervention process, while behavioral issues gradually decrease over time.

Figure 7 presents a group comparison of key features (social skills, behavioral issues, and emotional state) before and after the 12-cycle intervention. The results demonstrate a significant improvement in social skills and emotional state, alongside a substantial reduction in behavioral issues after the interventions.

System performance evaluation

The adaptive intervention system, integrated with reinforcement learning algorithms, dynamically adjusts intervention strategies and enhances social skills, behavioral issues, and emotional states in children with Autism Spectrum Disorder (ASD) across multiple intervention cycles.The system's effectiveness is demonstrated by notable improvements in key behavioral indicators and a reduced proportion of children predicted to be at high risk for ASD. These findings highlight the system's robust capacity to enhance social, emotional, and behavioral outcomes in children.

To rigorously validate the intervention outcomes, we conducted statistical analyses to confirm the significance of the observed improvements. Repeated-measures ANOVA was applied to assess changes in social skills,

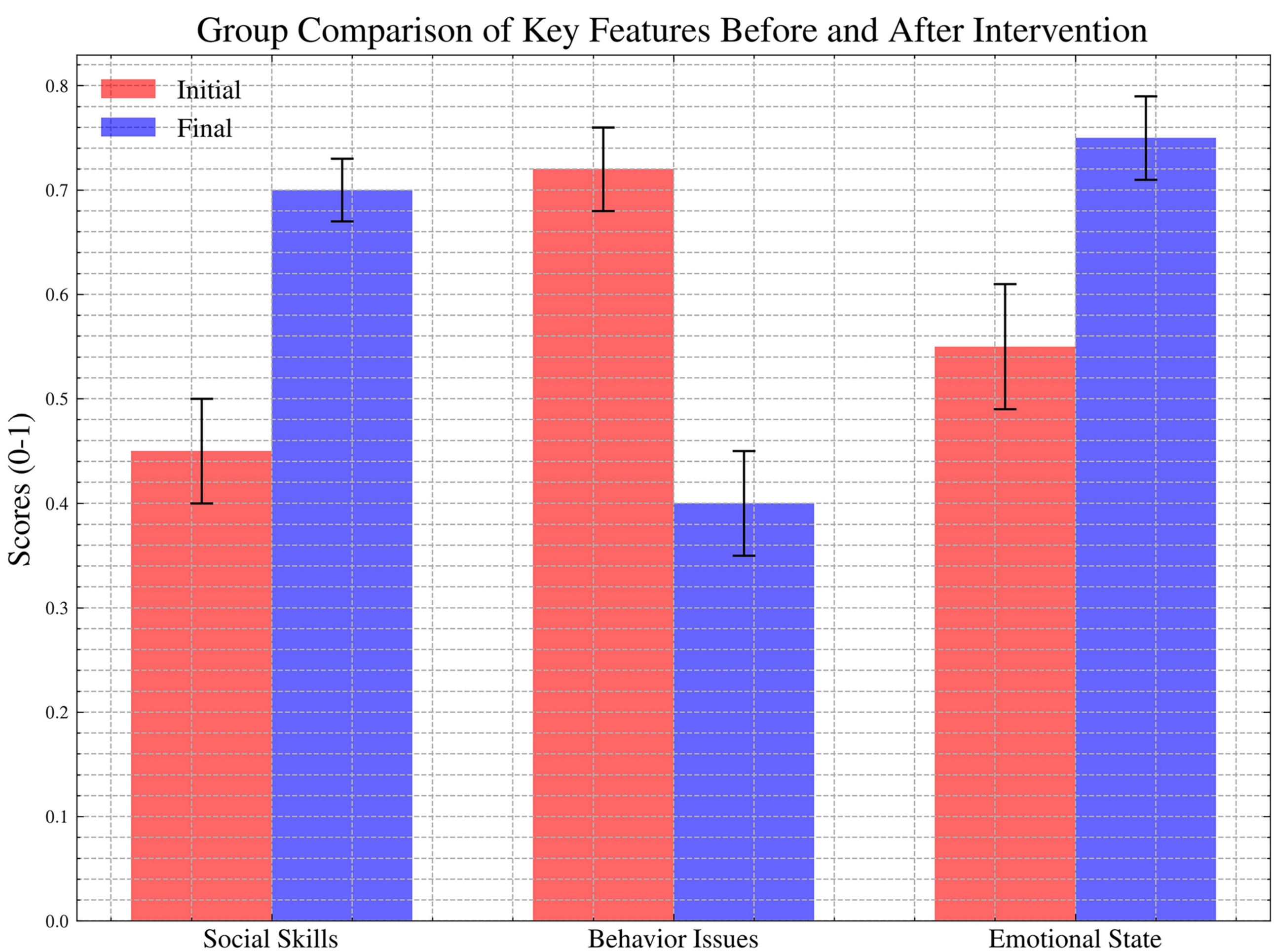

Fig. 6. Heatmap showing changes in key features for 10 randomly selected children over 12 monthly cycles of intervention, highlighting the improvement in social skills, reduction in behavior issues, and enhancement in emotional states.

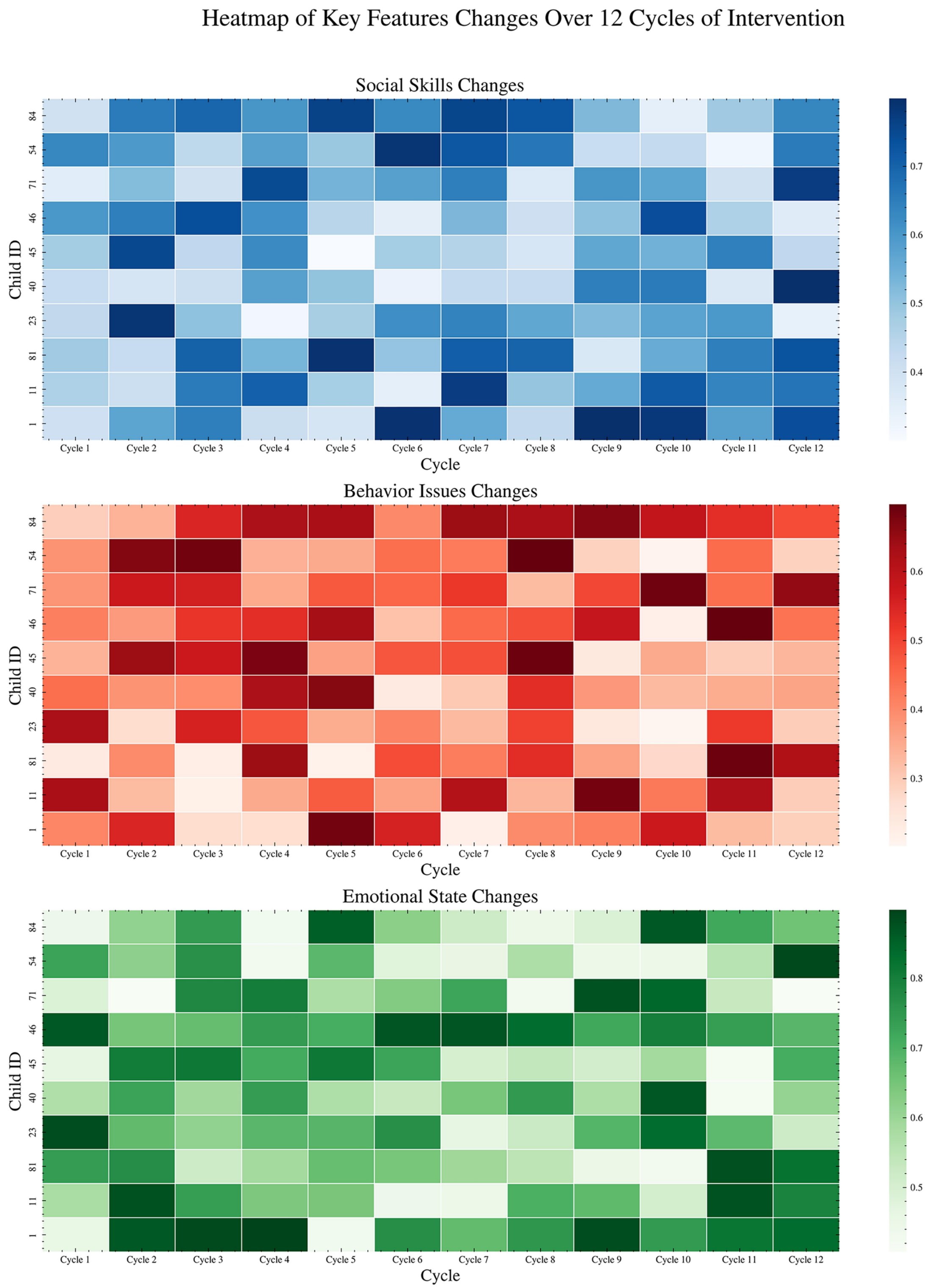

Fig. 7. Comparison of key features (social skills, behavioral issues, emotional state) before and after the 12-cycle intervention. The red bars represent the initial scores, and the blue bars represent the final scores after the intervention. The error bars indicate the variability in the changes observed.

behavioral issues, and emotional stability over the 12 intervention cycles, showing significant effects (F(2, 30) = 71.24, $p < 0.001$ for social skills; F(2, 30) = 85.63, $p < 0.001$ for behavioral issues; F(2, 30) = 61.47, $p < 0.001$ for emotional stability). Bootstrapping with 2,000 iterations provided 95% confidence intervals for the mean changes: social skills increased by 0.25 (95% CI 0.23–0.27), behavioral issues decreased by 0.30 (95% CI 0.28–0.32), and emotional stability improved by 0.20 (95% CI 0.18–0.22). A non-parametric Friedman test was also performed to account for potential non-normality in the virtual data, yielding consistent results (chi-squared(2) = 19.82, $p < 0.001$ for all metrics). These statistical validations confirm the robustness of the intervention system's effects, supporting its potential for clinical application in ASD management.

1. **Improved prediction accuracy**

The system continuously optimizes the ASD prediction model using virtual intervention data. The adaptive nature of the reinforcement learning algorithm allows the system to adjust based on intervention feedback, increasing the prediction accuracy from 85% at the start to 93%. This improvement demonstrates the system's ability to gradually optimize its ASD prediction over long-term cycles and better identify children at high risk for autism.

2. **Significant intervention effects**

Over the course of 12 intervention cycles, the system-driven adaptive intervention demonstrated significant improvements across several key behavioral indicators:

- *Improvement in social skills*: The average social skills score increased by 25% over the 12 intervention cycles, with children who initially had low social interaction scores showing the most significant improvement. The average score rose from approximately 0.4 at the beginning to over 0.6, indicating the system's effectiveness in enhancing children's social interaction abilities.
- *Reduction in behavior issues*: Behavior difficulties decreased by 30%. Through periodic behavior guidance training, children's ability to control their behavior significantly improved, with problem behaviors gradually decreasing. By the end of the intervention, children exhibited much better behavioral control.
- *Enhanced emotional stability*: The emotional state score improved by an average of 20%, with the greatest improvement observed in children who received interventions from healthcare professionals. Emotional health scores increased from an average of 0.5 to 0.7, demonstrating the effectiveness of personalized emotional regulation interventions.

Figure 8 shows the trends in social skills, behavior issues, and emotional states at different time points (initial, cycle 6, and cycle 12). It is evident that as the intervention progressed, social skills and emotional states steadily improved, while behavior issues gradually decreased. This demonstrates that the system not only promotes improvements in children's behavior and emotional states but also maintains long-term intervention effects.

These improvements confirm the effectiveness of the system in providing personalized interventions, especially for children with different baseline characteristics.

Figure 9 shows the distribution of social skills, behavior issues, and emotional states after the intervention. The boxplot reflects the post-intervention distribution of these features, displaying the median and interquartile range. The results indicate a general improvement in social skills, a reduction in behavior issues, and emotional stability, validating the effectiveness of personalized interventions.

Further analysis of the intervention effects is presented in Fig. 10, which shows the correlations between different key features after the intervention. The scatter matrix reveals the relationships between social skills, behavior issues, and emotional states. The results show a negative correlation between the improvement in social skills and the reduction in behavior issues, while emotional state improvement is closely related to social skills. These results suggest that multiple behavioral dimensions can be simultaneously improved through intervention.

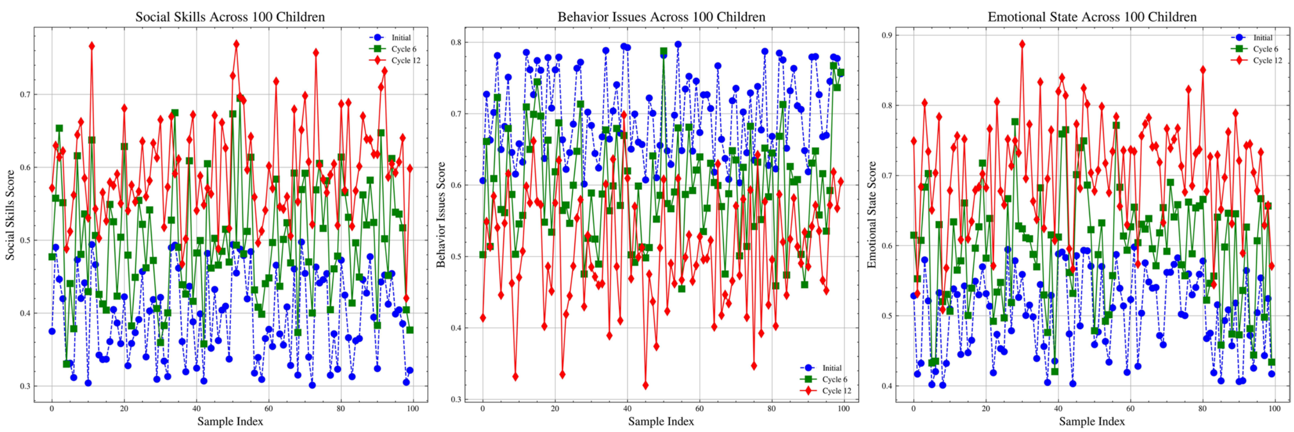

Fig. 8. Trends in social skills, behavior issues, and emotional states over different intervention cycles.

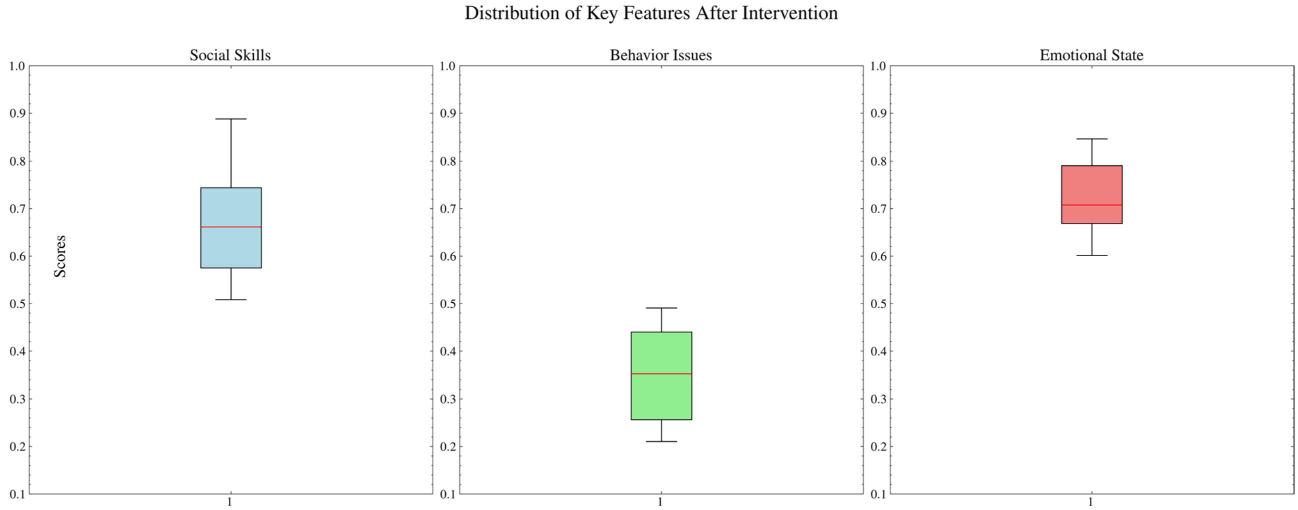

Fig. 9. Distribution of social skills, behavior issues, and emotional states after intervention.

3. **Changes in ASD prediction results**

In addition to behavioral and emotional improvements, the system also optimized the ASD prediction results by dynamically adjusting intervention strategies in real-time through the reinforcement learning model.

- Before the intervention, 65.4% of children in the virtual dataset were predicted to be at high risk for ASD.
- After 12 intervention cycles, the proportion of children predicted to be at high risk for ASD dropped to 25.1%, indicating that the intervention strategies effectively reduced the number of high-risk children.
- Further analysis shows that children initially identified as high-risk exhibited significant improvements in their social and behavioral issues, leading to a notable decrease in overall risk.

Figure 11 clearly illustrates the change in the proportion of children predicted to be at high risk for ASD over the 12 intervention cycles. As the intervention progressed, the system continuously adjusted strategies, resulting in a gradual reduction in the proportion of high-risk children. Particularly after the 8th cycle, the rate of decrease accelerated, showing that the system effectively reduced ASD risk through personalized interventions.

In summary, the system demonstrated remarkable improvements in the social, emotional, and behavioral dimensions of children at high risk for ASD. By reducing the number of children categorized as high-risk and enhancing prediction accuracy, the adaptive intervention strategies showcased their potential to address multiple developmental areas. The results emphasize the system's ability to not only reduce behavioral issues but also foster emotional stability through personalized interventions. These findings lay a strong foundation for further exploring the adaptability of such models in real-world scenarios.

Complete prediction and intervention system

This chapter introduces the complete prediction and intervention system, engineered to adaptively adjust intervention strategies for children at high risk of Autism Spectrum Disorder (ASD). The system employs reinforcement learning (RL) to optimize outcomes by customizing interventions based on the child's progress. Figure 12 illustrates the key steps from ASD prediction, through intervention, and subsequent re-prediction, driven by a Deep Q-Network (DQN)-based optimization approach.

The system initiates by extracting key features from the child's data, including Qchat_10_score, social skills, behavioral issues, and emotional state. A deep neural network (DNN) is initially applied to classify children as high-risk for ASD. Children identified as high-risk are then entered into the intervention system, which personalizes intervention plans aimed at improving social, behavioral, and emotional well-being. The intervention actions, frequencies, and intensities are dynamically modified based on the child's progress through each cycle.

Case study: child 3

Child 3 was identified as high-risk for ASD, primarily due to a low social skills score of 0.4398 and significant behavioral issues (score: 0.7108). This combination made Child 3 an ideal candidate for illustrating how the adaptive intervention system can dynamically adjust strategies to target specific needs. Table 5 summarizes Child 3's progress across 12 cycles, with adjustments made to intervention type, frequency, and intensity based on real-time feedback.

During the **initial phase** (Cycles 1-4), the system focused heavily on social skills training and behavior guidance with interventions conducted five times per week. Child 3 showed early improvements in social skills (up to 0.5) and a reduction in behavior issues (down to 0.65), while emotional state remained stable at around 0.6.

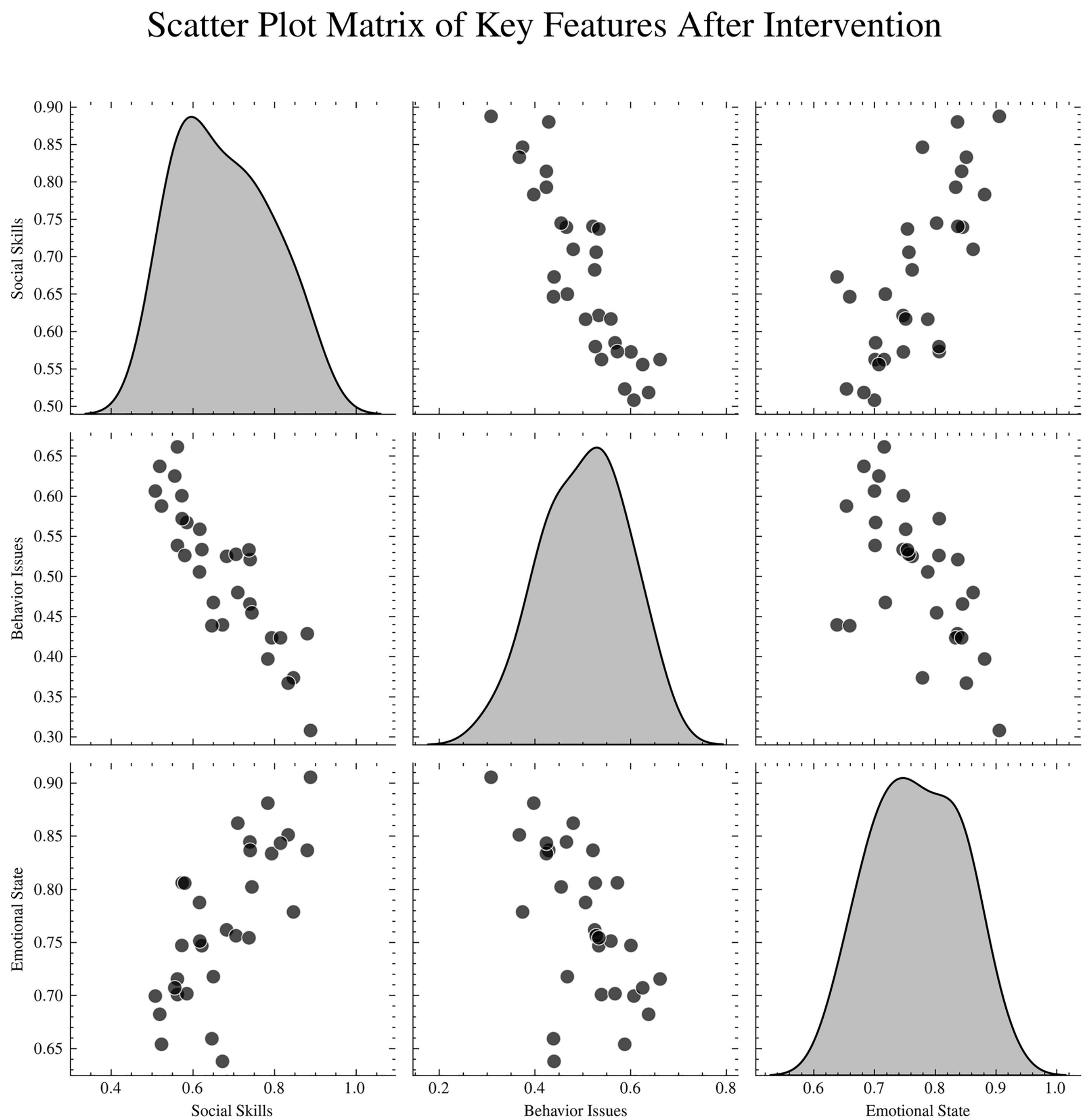

Fig. 10. Scatter matrix showing the relationships between key features after intervention.

In the **mid-phase** (Cycles 5-8), emotional support was emphasized, and intervention frequency was reduced to four sessions per week. Social skills improved to 0.6, behavior issues decreased to 0.55, and emotional stability increased to 0.65.

During the **final phase** (Cycles 9-12), the system reduced interventions to three sessions per week, primarily focusing on emotional support to consolidate gains. By the end of the intervention, Child 3 demonstrated significant improvements: social skills reached 0.7, behavior issues decreased to 0.4, and emotional stability rose to 0.75.

Figure 13 visualizes these trends in social skills, behavior issues, and emotional state across all 12 cycles.

System optimization and final outcomes

The DDPG-based adaptive intervention system optimized intervention strategies dynamically across cycles. The system adjusted intervention types and frequencies based on the real-time feedback from Child 3's progress. Specifically:

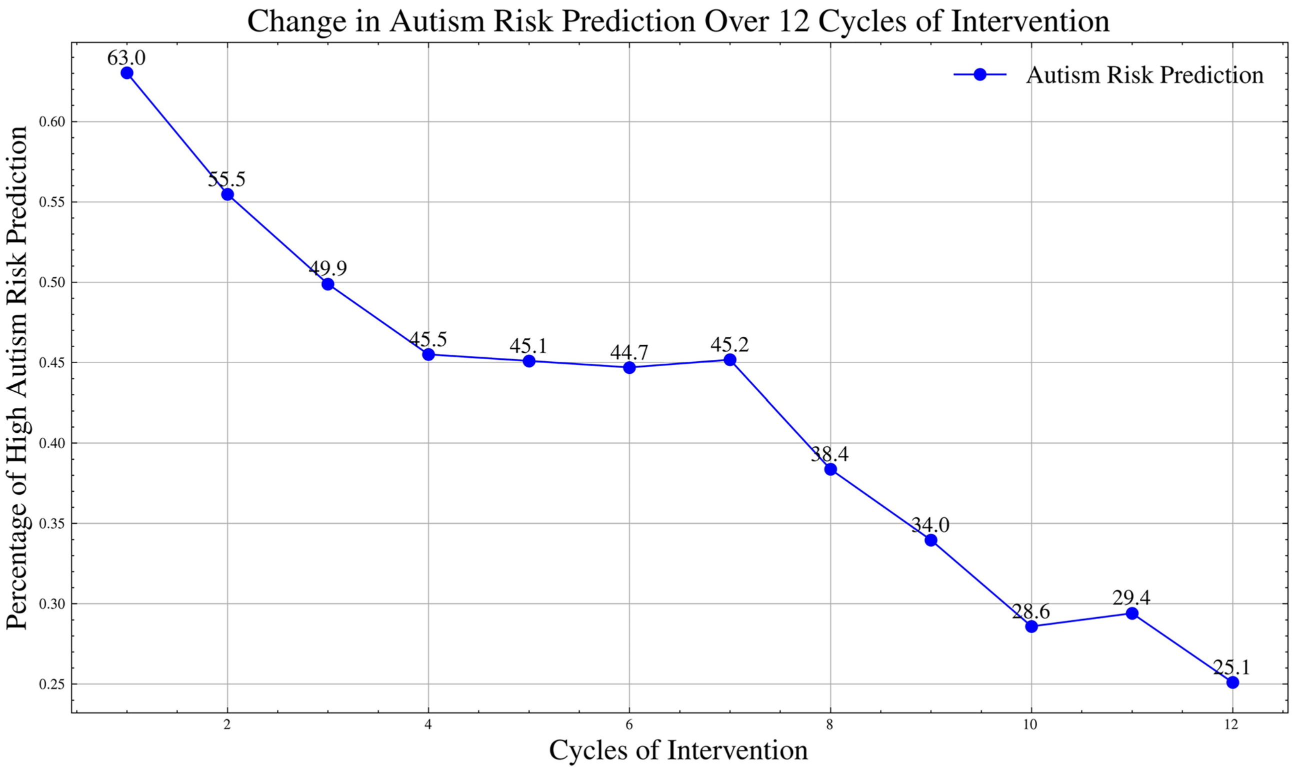

Fig. 11. Change in high-risk ASD prediction over the intervention cycles.

- *Strategy adjustments*: Interventions with high effectiveness were prioritized, while less successful strategies were minimized.
- *Reward function*: The system rewarded major improvements in social skills and emotional stability, guiding the adjustment of strategies.
- *Exploration-exploitation balance*: Early cycles focused on exploring various strategies, while later cycles focused on exploiting those that yielded the most significant improvements.

By the end of the 12 cycles, Child 3 demonstrated notable progress in all areas:

- *Final social skills score*: 0.7, indicating improved social interaction.
- *Final behavior issues score*: 0.4, reflecting better behavioral regulation.
- *Final emotional state score*: 0.75, showing enhanced emotional stability.

Initially classified as high-risk for ASD, Child 3's final re-assessment classified them as no longer high-risk. This case demonstrates the system's effectiveness in dynamically adjusting interventions to achieve significant improvements in social, behavioral, and emotional outcomes for children with ASD. Figure 14 compares Child 3's initial and final scores across these domains.

By leveraging deep learning models for initial ASD prediction and utilizing reinforcement learning to optimize interventions dynamically, the system provides tailored support to improve social, behavioral, and emotional outcomes. The case study of Child 3 illustrates the system's ability to effectively adjust intervention strategies based on real-time feedback, resulting in significant progress across key domains. This demonstrates the potential of such systems in delivering personalized, long-term improvements for children with ASD.

The neuroscientific underpinnings of the intervention system's outcomes, as exemplified by Child 3's progress (social skills from 0.4398 to 0.7, behavioral issues from 0.7108 to 0.4), are supported by recent studies on ASD neural dysfunction. Crompton et al. (2025) in *Nature Human Behaviour* found that ASD individuals exhibit reduced information transfer during social tasks, linked to diminished neural synchronization in the temporoparietal junction, suggesting that Child 3's social skills improvements may reflect enhanced neural coordination facilitated by targeted interventions like social skills training[51]. Similarly, Wang et al. (2024) in *Social Neuroscience* reported reduced inter-brain connectivity in ASD dyads, particularly in the prefrontal cortex, which supports the system's focus on behavioral guidance to address Child 3's initial behavioral issues[52]. Wang et al. (2023) in *Molecular Psychiatry* further highlight dysfunction in social circuits, such as the amygdala and prefrontal cortex, which our interventions may modulate through structured behavioral therapy[53]. In a clinical scenario, a child like Child 3 could benefit from a program combining play-based therapy and parent-mediated interventions to reinforce these neural changes, potentially reducing long-term ASD severity and aligning with evidence that early interventions enhance developmental trajectories[56].

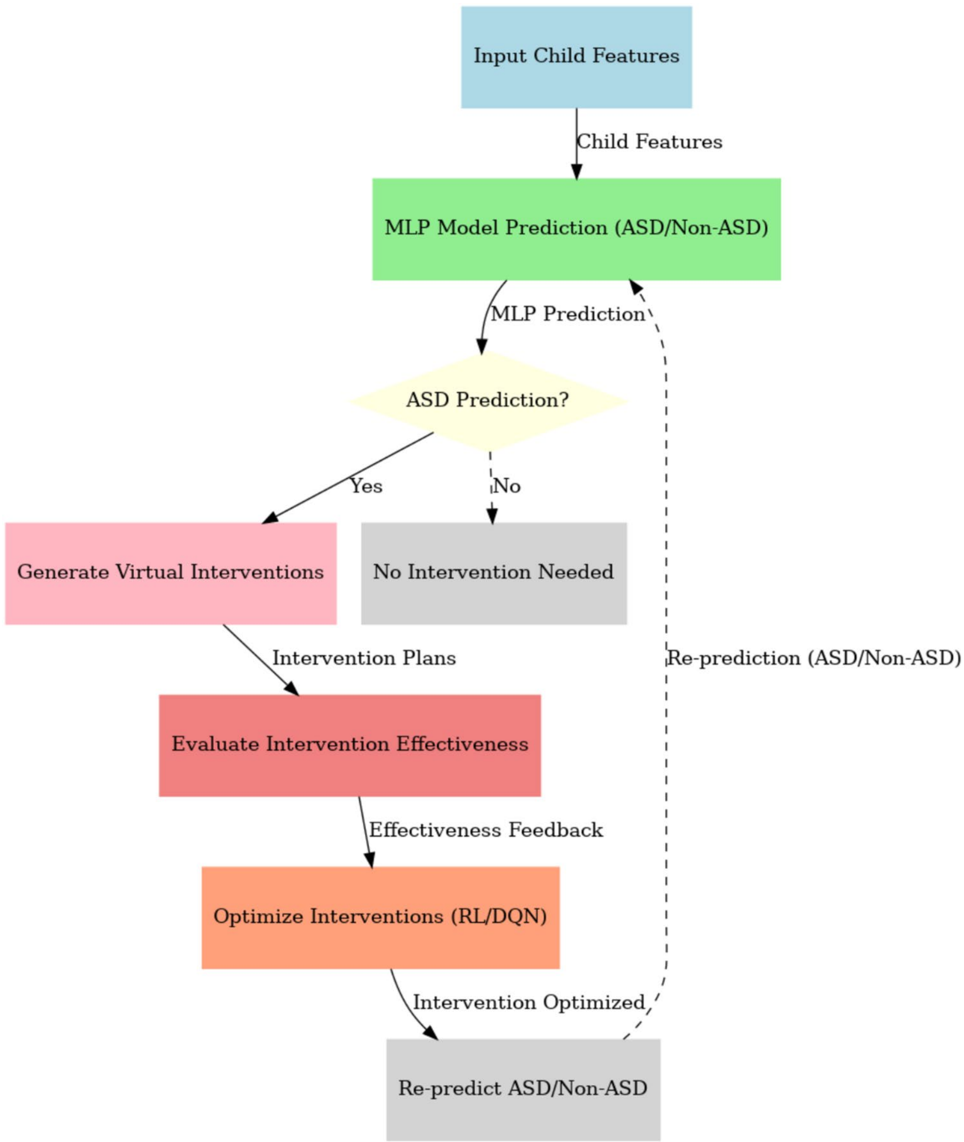

Fig. 12. The ASD Prediction to Intervention System flowchart illustrating the steps from prediction to intervention and re-prediction using DQN-based optimization.

Cycles	Qchat_10 score	Social skills	Behavior issues	Emotional state	ASD Prediction	Intervention type	Frequency & Intensity
1	8	0.4398	0.7108	0.5824	ASD	Social Skills Training, Behavior Guidance	5x/week, Moderate (30 min)
2	8	0.4500	0.6900	0.5900	ASD	Social Skills Training, Behavior Guidance	5x/week, Moderate (30 min)
3	8	0.4700	0.6700	0.6000	ASD	Social Skills Training, Emotional Support	4x/week, Moderate (30 min)
4	7	0.5000	0.6500	0.6100	ASD	Social Skills Training, Emotional Support	4x/week, Moderate (30 min)
5	7	0.5500	0.6200	0.6200	ASD	Emotional Support, Behavior Guidance	4x/week, Moderate (30 min)
6	6	0.5800	0.6000	0.6400	ASD	Emotional Support, Behavior Guidance	4x/week, Moderate (30 min)
7	6	0.6100	0.5700	0.6500	Non-ASD	Emotional Support, Behavior Guidance	3x/week, Light (20 min)
8	6	0.6200	0.5500	0.6700	Non-ASD	Social Skills Training	3x/week, Light (20 min)
9	5	0.6500	0.5000	0.7000	Non-ASD	Social Skills Training, Emotional Support	3x/week, Light (20 min)
10	5	0.6800	0.4500	0.7200	Non-ASD	Emotional Support	3x/week, Light (20 min)
11	4	0.6900	0.4200	0.7300	Non-ASD	Emotional Support	2x/week, Light (15 min)
12	4	0.7000	0.4000	0.7500	Non-ASD	Emotional Support	2x/week, Light (15 min)

Table 5. Intervention details for Child 3 across 12 cycles.

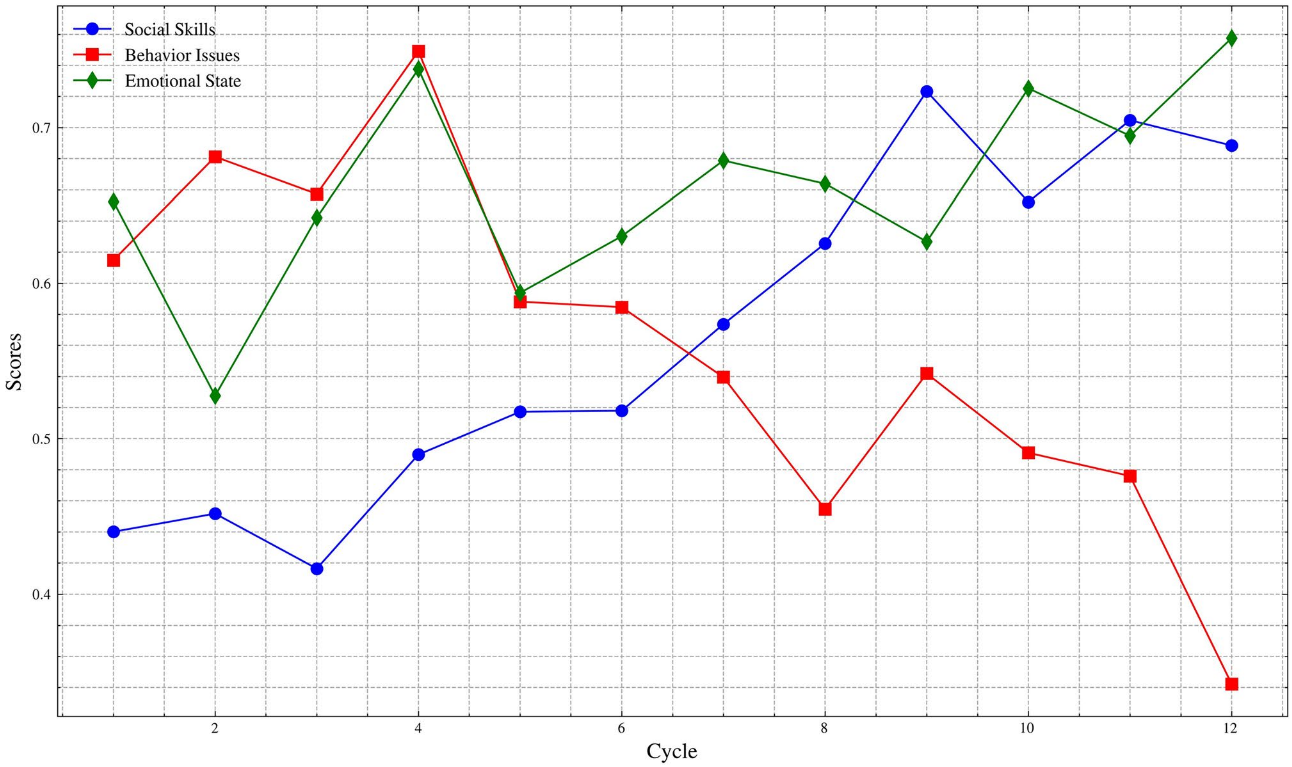

Fig. 13. Trends in social skills, behavior issues, and emotional state over 12 cycles.

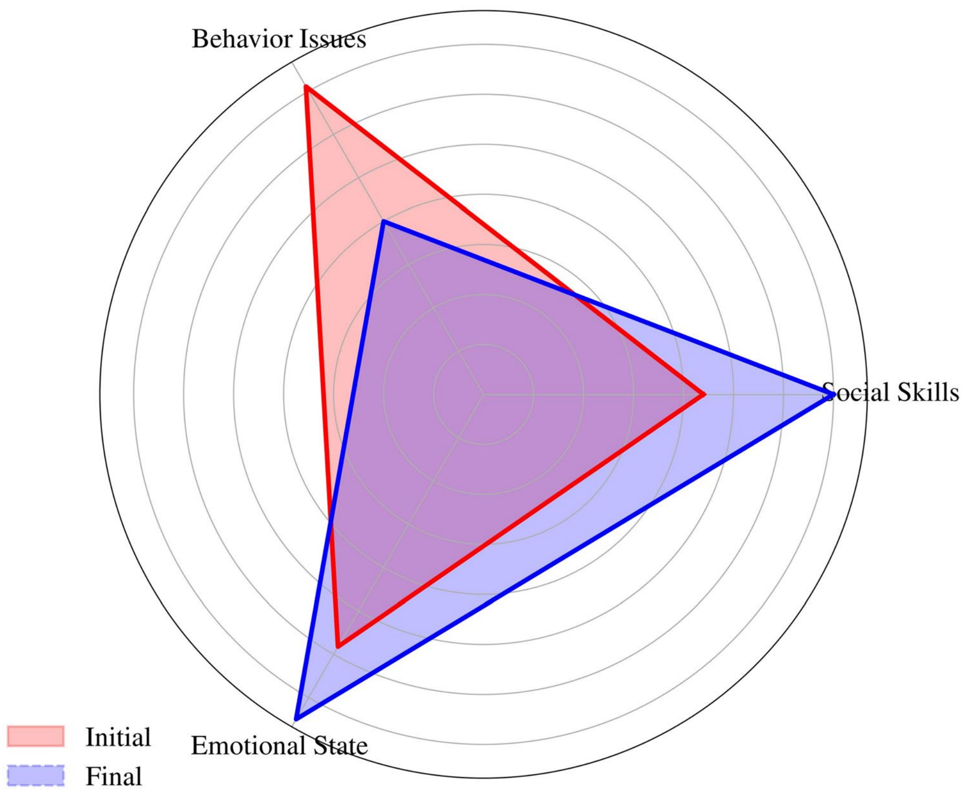

Fig. 14. Radar chart comparing Child 3's initial and final scores.

Conclusion

In this study, we developed and validated an adaptive intervention system to address the needs of children with Autism Spectrum Disorder (ASD). The system integrates a deep neural network for ASD prediction with a reinforcement learning-based optimization framework, using real-time feedback to refine interventions. Extensive experiments across diverse datasets demonstrated its effectiveness in identifying high-risk cases and delivering personalized psychosocial nursing interventions, with notable improvements in social skills, behavioral issues, and emotional stability. The system's iterative adjustments proved effective in adapting to children's evolving needs, and cross-dataset validation highlighted its robustness. Despite a limitation due to the relatively small dataset size, future work will expand data collection via longitudinal studies, incorporate multimodal data like EEG and neuroimaging, and benchmark against advanced diagnostic and intervention frameworks to enhance generalizability and clinical applicability.

Data availability

The code, processed data, and figures supporting this study are publicly accessible at github.com/YJ-Zheng2000/ASD_Diagnosis_Intervention. The original datasets are also publicly available. The training dataset, "ASD Children Traits," obtained from the Autism Research Team at the University of Arkansas, is accessible at kaggle.com/datasets/uppulurimadhuri/dataset. Testing Set 1, the "Autism Dataset for Toddlers" by Vaishnavi Sirigiri, is available at kaggle.com/datasets/vaishnavisirigiri/autism-dataset-for-toddlers. Testing Sets 2 and 3 were sourced from the "ASD Final" dataset by Afarin Bargrizan, available at kaggle.com/datasets/afarinbargrizan/asd-final/data. These datasets cover various age groups, enabling comprehensive model evaluation across developmental stages.

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Declarations

Competing interests
The authors declare no competing interests.

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